Relational Calculus is a procedural query language.

Answer:

- True
- False

Normalization is the process of decomposing unsatisfactory "bad" relations.

Answer:

- (●) True
- ( ) False

Primary key is a minimal of the Super key.

Answer:

- True
- False

What Should an Entity Be?

Choose at least one answer.

- [ ] a. A user of the database system
- [x] b. An object that will be composed of multiple attributes
- [ ] c. An output of the database system
- [x] d. An object that will have many instances in the database

What are the Characteristics of **Bad** Database?
Select all that applies.

Choose at least one answer.

- [ ] a. Having more comples relationships among the data
- [x] b. Would contain inconsistent and invalid data
- [x] c. Would require excessive effort to change
- [x] d. Not satisfy all user requirements

What are the Characteristics of **Bad** Database?
Select all that applies.

Choose at least one answer.

- [ ] a. Having more comples relationships among the d
- [x] b. Would contain inconsistent and invalid data
- [x] c. Would require excessive effort to change
- [x] d. Not satisfy all user requirements

Why don't we need to consider of using traditional file system? Select all the answers which appli

Choose at least one answer.

- [ ] a. Neet to have a high security on data
- [x] b. Redundancy in defining and storing data
- [x] c. Wasted storage space
- [x] d. Redundant efforts to maintain common data up-to-date

B19_IN 1400 - Fundamentals of Databases and Database Design

Question **1**

Not yet answered

Marked out of 1.00

Flag question

Which of the following is/are possible candidate key(s) for Customer relation?

Customer (CustomerNo, CustomerName, PostalCode, Address)

Select one:

- a. CustomerNo only
- b. CustomerNo, PostalCode only
- c. PostalCode, Address only
- d. CustomerName only

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Time left **0:44:15**

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Question **2**

Not yet answered

Marked out of 1.00

Flag question

Which of the following is/are database management system(s)?

A. MS Access

B. MS Excel

C. MySQL

Select one:

- a. B only
- b. C only
- c. A, B only
- d. A, C only

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Time left **0:42:22**

B19_IN 1400 - Fundamentals of Databases and Database Design

Question **4**

Not yet answered

Marked out of 1.00

Flag question

Which of the following statement is true regarding user-data in a database developed to store data of a university?

Select one:

- a. It stores details of each student and lecturer.
- b. It describes users of the database and their privileges.
- c. It stores outliers in the student details.
- d. It describes and maintains the structure of the database.

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Time left **0:38:45**

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Question **7**

Not yet answered

Marked out of 1.00

Flag question

Consider the following relations to answer the question below.

Order (OrderId, CustomerNo, OrderedDate)

Customer (CustomerNo, Email, CustomerName, PostalCode, Address)

There is a *has* relationship between Customer relation and Order relation.

Cardinality of the *has* relationship between Customer and Order relation:

Select one:

- (selected) a. One to many
- b. Many to one
- c. One to one
- d. Many to many

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Time left **0:31:57**

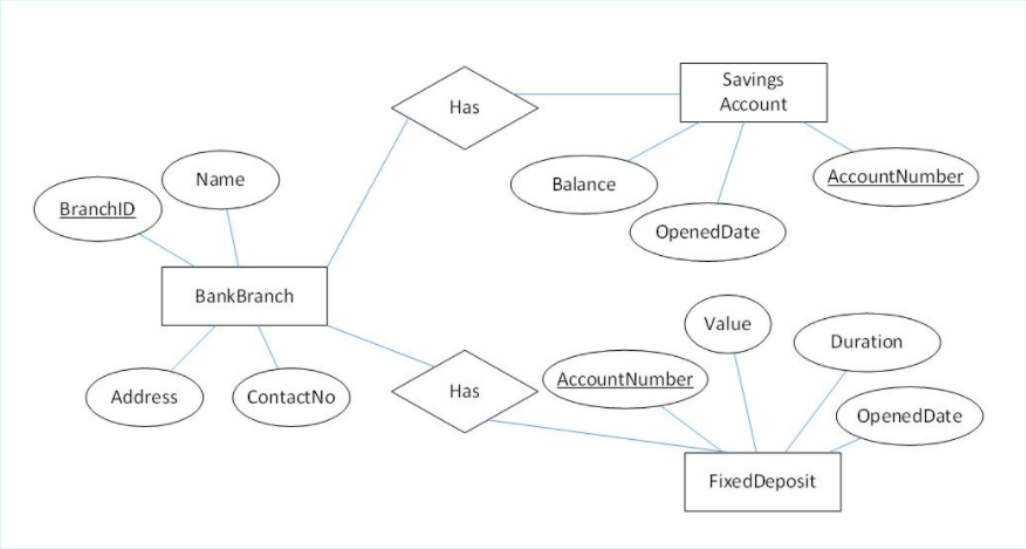
Question **8**

Not yet answered

Marked out of 1.00

Flag question

The following Entity Relationship (ER) diagram has been converted to Enhanced ER (EER) diagram.



Identify the most suitable modification that could be done when converting to an EER diagram.

Select one:

- (●) a. Created a new entity called Account and included common attributes of SavingsAccount and FixedDeposit in Accounts entity. Additional attributes are kept in SavingsAccount and FixedDeposit entities.
- ( ) b. Included common attributes of SavingsAccount and FixedDeposit in BankBranch.
- ( ) c. Created a new entity as Account and included all the attributes of SavingsAccount and FixedDeposit in Accounts entity. Removed SavingsAccount and FixedDeposit entities.
- ( ) d. Included a ternary relationship between BankBranch, SavingsAccount, and FixedDeposit.

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Question **9**

Not yet answered

Marked out of 1.00

Flag question

Type of attribute that is best suited to represent the following attributes for Vaccination entity respectively are:

ManufacturedDate, ExpiryDate, DaysToExpiration

Select one:

- a. Stored attribute, derived attribute, stored attribute
- b. Stored attribute, stored attribute, derived attribute
- c. Derived attribute, stored attribute, stored attribute
- d. Stored attribute, stored attribute, stored attribute

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Question **10**

Not yet answered

Marked out of 1.00

Flag question

A tuple in a relationship is:

Select one:

- a. A relational table.
- b. A set of attributes in an entity.
- c. An instance of an entity.
- d. A field of an entity.

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Question **17**

Not yet answered

Marked out of 1.00

Flag question

Which of the following is a main function of a database administrator?

Select one:

- a. Write application programs.
- b. Define security and integrity rules of a database.
- c. Use application programs written by another person.
- d. Use a database query language to form their requests.

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Question **18**

Not yet answered

Marked out of 1.00

Flag question

Consider the following relations to answer the question below.

Hotel (HotelNo, HotelName, Location, OpenedDate)

Booking (HotelNo, GuestNo, RoomNo, BookedDate, LastUpdatedDate)

Guest (GuestNo, GuestName, Email, Address)

Some of the hotels are not booked by guests yet. There is a *has* relationship between Hotel and Booking relation.

What is/are the suitable participation constraint(s) to describe the above situation?

Select one:

- a. There is partial participation of Booking entity in 'has' relationship.
- b. There is total participation of Hotel entity in 'has' relationship.
- c. There is total participation of Booking entity in 'has' relationship.
- d. There is partial participation of Booking with Hotel relation.

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Question **19**

Not yet answered

Marked out of 1.00

Flag question

Identify the super key of the following relation. The primary key is underlined.

Item(ItemName, ItemType, ExpiryDate, Description)

Select one:

- a. ItemType
- b. ItemType, ExpiryDate
- c. ItemName, ExpiryDate
- d. ItemType, ExpiryDate, Description

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Question **2**

Not yet answered

Marked out of 1.00

Flag question

Which of the following statement(s) will insert a new record to the Student table?

Student (StudentId, StudentName, City, District)

A. INSERT INTO Student VALUES (102, 'Sarath', 'Borella', 'Colombo');

B. INSERT INTO Student (StudentId, StudentName, District, City) VALUES (102, 'Sarath', 'Colombo', 'Borella');

C. INSERT INTO Student VALUES (102, 'Sarath', 'Colombo');

Select one:

- a. B only
- b. A and C only
- c. A only
- d. A and B only

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Time left **0:39:23**

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Question **3**

Not yet answered

Marked out of 1.00

Flag question

Identify the number of students in each district for the Student relation. However, only consider districts with more than 10 students.

Student (StudentId, StudentName, City, District)

Select one:

- a. SELECT District, COUNT(StudentId)
  FROM Student
  HAVING COUNT(StudentId) > 10;
  GROUP BY District
- b. SELECT District, COUNT(StudentId) >10
  FROM Student
  GROUP BY District
- c. SELECT District, COUNT(StudentId)
  FROM Student
  WHERE COUNT(StudentId) > 10
  GROUP BY District
- d. SELECT District, COUNT(StudentId)
  FROM Student
  GROUP BY District
  HAVING COUNT(StudentId) > 10;

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Question **7**

Not yet answered

Marked out of 1.00

Flag question

Which of the following SQL statement is used to modify District from Colombo (current value) to Kandy (new value) in the Student table?

Student (StudentId, StudentName, City, District)

Select one:

- a. UPDATE Student SET District = 'Colombo' WHERE District = 'Kandy';
- b. UPDATE Student WHERE District = 'Colombo' SET District = 'Kandy';
- c. UPDATE Student WHERE District = 'Kandy' SET District = 'Colombo';
- d. UPDATE Student SET District = 'Kandy' WHERE District = 'Colombo';

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Question **8**

Not yet answered

Marked out of 1.00

Flag question

A user needs to change the number of the rows of the Student table by adding details of a new *Batch*.

Select the most suitable SQL command type.

Select one:

- a. Data Definition Language
- b. Data Manipulation Language
- c. Data Control Language
- d. Data Query Language

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